



Department of Computer Science and Engineering

Jashore University of Science and Technology

2nd Year 1st Semester Final Examination-2023

Session: 2020-21

Course Code: HUM 2101

Course Title: Sociology and Disaster Management

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

1. a) What do you understand by the term 'Government'? What is the difference between Absolute monarchy and Constitutional monarchy? 2+2

b) In a feudal society, states is based on land ownership. What is the political importance of feudalism? 04

c) What are the organs of the government? Differentiate among them. 1+3

2. a) Define the following terms: 1×4=4

i) Budget
ii) Hearing

iii) Lobby
iv) Amendment

b) What do you understand by the term 'Modern State'? What are the functions of the modern state? 2+2

c) What are the contemporary aspects of public administration regarding the following topics: 2+2

i) Protecting and advancing democracy
ii) Ensuring Environmental Sustainability

3. a) What do you mean by social dynamics? 03

b) Juvenile gangs are the result of social unrest. Give your opinion. 04

c) What are the impacts of technological factors in real life disaster management risk reduction? 05

4. a) What do you understand by the term 'Sociology'? What are the significant features of sociology? 2+2

b) What is 'Social Structure'? To understand the social structure, we have to understand different components. What are those components of social structure? 1+3

c) Define following terms: 1×4=4

i. Globalization
ii. Bureaucracy

iii. Caste
iv. Human Ecology

5. a) Differentiate between: 2+2

i.) Quantitative data and Qualitative data
ii.) Open-ended questions and close-ended questions.

b) What do you understand by the term 'Research'? To do any kind of social research, you have to identify a study area. What kinds of issues do you have to consider before selecting an area as you study area for research? 1+2

P.T.O

7.

Consider an algorithm as follows:

```

Bellman-Ford(G, w, s)
// Graph  $G=(V, E)$ ,  $w: V \rightarrow R$  a weight function,  $s$  is the
source vertex
// uses distance estimates  $d[v]$  for  $v$  in  $V$ 
// and parents  $p[v]$  for  $v$  in  $V$  with  $p[s]=s$  for the source
1 Initialize(G, s)
2 for  $i = 1$  to  $|V|-1$ 
3   for each edge  $(u, v)$  in  $E$  do
4     Relax( $u, v, w$ )
5   end
6 end
7 for each edge  $(u, v)$  in  $E$  do
8   if  $d[v] > d[u] + w(u, v)$  then return false
9 end
10 return true

```

Famim

- a) Give the code of the procedure Relax(u, v, s). 05
- b) Derive the complexity of Bellman-Ford, assuming that Initialize(G, s) takes $\Theta(V)$ time. 04

Note: Do not just state the result, but explain what each loop contributes. Use line numbers as references.

- c) What does the loop in lines 7–9 do? Explain in brief. 03

8.

Consider a weighted directed graph with 5 vertices and weights on the edges:

$w(1, 2) = 3$, $w(1, 3) = 1$, $w(2, 3) = 3$, $w(2, 4) = 2$,
 $w(2, 5) = -2$, $w(3, 5) = 5$, $w(4, 2) = 1$, $w(4, 5) = -3$

- a) Compute the minimum cost path from 1 to 5. 04
- b) Explain why Dijkstra's algorithm does not work for this type of problem. 03
- c) What kind of algorithm will you use? Justify your opinion. 05

Illustrate the execution of the algorithm on this graph.

-----The End-----



Jashore University of Science and Technology
Department of Computer Science and Engineering

Semester Final Examination-2023

2nd Year 1st Semester, Session: 2020-2021

Course Code: CSE 2107

Course Title: Numerical Analysis

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Write short notes on-

1. Accuracy
2. Precision
3. Round-off errors
4. Floating-point representation

4*2

b) Determine the range of integers in base 10 that can be represented on a 32-bit computer.

04

2. a) Explain the concept of

1. Convergence
2. Divergence.

2*2

b) Determine the root of $f(x) = 0.95x^3 - 5.9x^2 + 10.9x - 6$

4+4

1. Using Newton-Raphson method (three iterations, $x_i = 3.5$)

2. Using the second method (three iterations, $x_{i-1} = 2.5$ and $x_i = 3.5$)

3. a) Determine the real roots of $f(x) = -0.5x^2 + 6.25x + 4.5$ using three iterations of bisection method, initial guesses of $x_l = 5$ and $x_u = 10$. Calculate the estimated error and true error after each iteration.

06

b) Describe the False-position-method with proper figures.

04

c) Differentiate the Bisection and False-position methods.

02

4. Suppose, you are new in Jashore. You have gathered the following information relating the housing rent of three-bedroom homes with the sizes located nearly at Jashore Boat Club.

Size (in Square Feet)	1800	2100	2400	2200
Housing Rent (in Taka)	13200	15400	17500	14800

a) Considering the above data, explicitly write Lagrange interpolation formula. Also estimate the rent for a house of 2000 square feet (using Lagrange formula).

06

b) Considering the given data, write a C program by implementing the Lagrange interpolation that takes the size of a house as input and prints the estimated housing rent as output.

06

5. Use Muller's method to determine the real roots of $f(x) = x^3 + x^2 - 3x - 5$.

06

P.T.O

- Q11) What are the implications of the following research methods: 2+2
- Ethnography
 - Participant observation
- Q12) What is the meaning of 'Community' to you? 01
- Q13) Culture is the systems of knowledge shared by a relatively large group of people. But sometimes, cultural lag happens. What do you understand by the term 'cultural lag'? 02
- Q14) What does a society become complex? State this aspect mentioning the components of a complex society. 1+2
- Q15) What is capitalism? What do you know about the growth of capitalism between the time 1600 and 1700s? 1+3
- Q16) Differentiate between material and non-material culture. 03
- Q17) "Nuclear families have a functional fit with industrial society"- Justify the statement. 03
- Q18) What are the relationships between industrial society and the development of urbanization? 03
- Q19) Sketch the following framework: 2+2
- Disaster Management institutional framework
 - Disaster Management Regulatory framework
- Q20) What are the difference between the nuclear family and the extended family? 02
8. a) The population is the whole number of inhabitants occupying an area and continually being modified by increases and losses. What are the factors which affect population change? 03
- b) What are pull and push factors of migration. 02
- c) What is the future of technology in disaster management? 03
- d) What types of technologies in disaster risk reduction can be used to ensure more effective and creative disaster management? 04

-----The End-----



Department of Computer Science and Engineering
Jashore University of Science & Technology
2nd Year 1st Semester (2020-21) - 2023

Course Code: MATH 2101

Total Time: 3.00 hours

Course Title: Matrices and Vectors

Total Marks: 72.0

(Note: Answer any six (06) set of questions from the following eight (08) set of questions)

- 1 (a) Describe scalar and vector fields with examples. Determine the angles α, β and γ which the vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ makes with the positive directions of the coordinate axes and show that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$. 05
- (b) Define rectangular unit vectors. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are non-coplanar vectors determine whether the vectors $\mathbf{r}_1 = 2\mathbf{a} - 3\mathbf{b} + \mathbf{c}$, $\mathbf{r}_2 = 3\mathbf{a} - 5\mathbf{b} + 2\mathbf{c}$, and $\mathbf{r}_3 = 4\mathbf{a} - 5\mathbf{b} + \mathbf{c}$ are linearly independent or dependent. 04
- (c) If $\mathbf{r}_1 = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, $\mathbf{r}_3 = -2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, and $\mathbf{r}_4 = 3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$, find scalars a, b, c such that $\mathbf{r}_4 = a\mathbf{r}_1 + b\mathbf{r}_2 + c\mathbf{r}_3$. 03
- 2 (a) Find an equation for the plane determined by the points $P_1(2, -1, 1)$, $P_2(3, 2, -1)$, and $P_3(-1, 3, 2)$. 03
- (b) Define reciprocal sets of vectors. If $\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$, $\mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$ and $\mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$; prove that $\mathbf{a} = \frac{\mathbf{b}' \times \mathbf{c}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}$, $\mathbf{b} = \frac{\mathbf{c}' \times \mathbf{a}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}$, $\mathbf{c} = \frac{\mathbf{a}' \times \mathbf{b}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}$. 05
- (c) Determine a unit vector perpendicular to the plane of $\mathbf{A} = 2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$. 04
- 3 (a) Define continuity and differentiability. A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. 05
- (b) If $\mathbf{A} = \sin u\mathbf{i} + \cos u\mathbf{j} + u\mathbf{k}$, $\mathbf{B} = \cos u\mathbf{i} - \sin u\mathbf{j} - 3\mathbf{k}$, and $\mathbf{C} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, find $\frac{d}{du}(\mathbf{A} \times (\mathbf{B} \times \mathbf{C}))$ at $u = 0$. 03
- (c) If $\mathbf{A} = (2x^2y - x^4)\mathbf{i} + (e^{xy} - y\sin x)\mathbf{j} + (x^2\cos y)\mathbf{k}$, find: $\frac{\partial A}{\partial x}, \frac{\partial A}{\partial y}, \frac{\partial^2 A}{\partial x^2}, \frac{\partial^2 A}{\partial y^2}, \frac{\partial^2 A}{\partial x \partial y}, \frac{\partial^2 A}{\partial y \partial x}$. 04
- 4 (a) Find equations for the tangent plane and normal line to the surface $z = x^2 + y^2$ at the point $(2, -1, 5)$. 04
- (b) If $\nabla U = 2r^4\mathbf{r}$, find U . Prove that the vector $\mathbf{A} = 3y^4z^2\mathbf{i} + 4x^3z^2\mathbf{j} - 3x^2y^2\mathbf{k}$ is solenoidal. 04
- (c) Find the acute angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point $(1, -2, 1)$. 04
- 5 (a) State the Green's theorem. 02
- (b) If $\vec{F} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}$, evaluate $\iiint_V (\nabla \times \vec{F}) dV$, where V is the closed region bounded by $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$. 06
- (c) Find the angle between the vectors $\vec{i} + \vec{j} + \vec{k}$ and $\lambda^2\vec{i} - 2\lambda\vec{j} + \vec{k}$. For what value of λ will the vectors be perpendicular? 04

P.T.O

b) Divide the polynomial $f(x) = x^5 - 5x^4 + x^3 - 6x^2 - 7x + 10$ by the monomial factor $x - 2$. 06

6. a) Evaluate the polynomial $y = x^3 - 7x^2 + 8x - 0.35$ at $x = 1.37$. Use 3-digit arithmetic with chopping. Evaluate the percent relative error. 06

b) Explain the Taylor-series with proper examples. How we can use Taylor-series to estimate truncation error. 06

7. a) Consider the following system of equations. 06

$$\begin{array}{rrcr} 1x_1 & + & 3x_2 & + & 2x_3 & = & 2 \\ 2x_1 & + & 7x_2 & + & 7x_3 & = & -1 \\ 2x_1 & + & 5x_2 & + & 2x_3 & = & 7 \end{array}$$

Apply Gauss elimination method to solve the given system. Also comment on the nature of the solution.

b) Now consider the new system of linear equations. Solve the following system and comment on the nature of the solution by applying Gauss elimination method. 06

$$\begin{array}{rrcr} 1z_1 & + & 3z_2 & + & 2z_3 & = & 4 \\ 1z_1 & + & 2z_2 & + & 1z_3 & = & 7 \\ 2z_1 & + & 5z_2 & + & 3z_3 & = & 10 \end{array}$$

8. a) Solve the following system of equations by LU decomposition. 08

$$8x_1 + 4x_2 - x_3 = 11$$

$$-2x_1 + 5x_2 + x_3 = 4$$

$$2x_1 - x_2 + 6x_3 = 7$$

b) Determine the matrix inverse from (8a). 04

$$\begin{array}{l} x_1 = 1 \\ x_2 = 9 \\ x_3 = -12 \end{array}$$



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2nd Year 1st Semester Final Examination-2023

Session: 2020-21

Course Code: CSE 2103

Course Title: Algorithm Analysis and Design

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

1. a) Why algorithm study is important? What are the steps to develop an algorithm? What are the characteristics of a good algorithm. 2+2+2

b) Elaborate the idea of best case, average case and worst-case scenario of an algorithm with examples. What are the corresponding notions 06

2. a) Sort the following items in descending order using merge sort and quick sort 500, 600, 100, 23, 152, 712, 900 3+3

b) Identify the space and time complexity for your responses in question 2(a). 06

3. a) Design an algorithm for fractional knapsack problem. Specify at least two shortcomings of your algorithm. What is the space complexity of your problem? 3+2+1

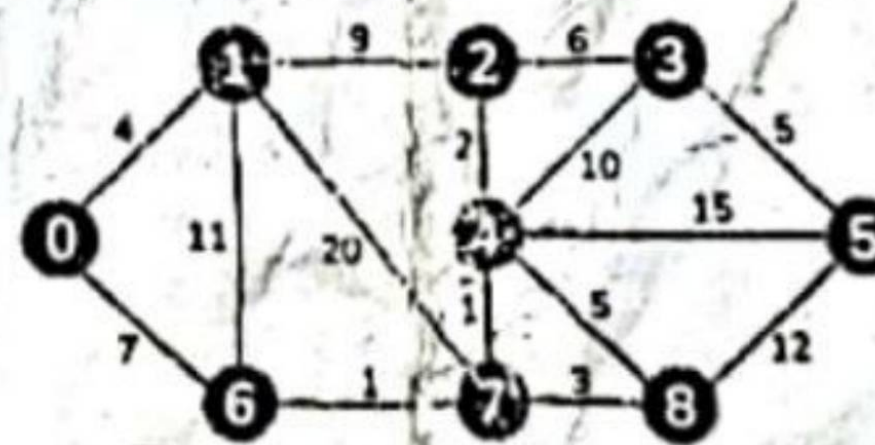
b) Discuss the max-min algorithm with its computational advantages over the linear search. 4+2

4. a) Devise the comparative study between dynamic programming and greedy solution 04

b) Identify the lowest cost and corresponding sequence for the following matrix chain 10, 15, 2, 5, 15. 6+2

Discuss the time complexity of the above algorithm

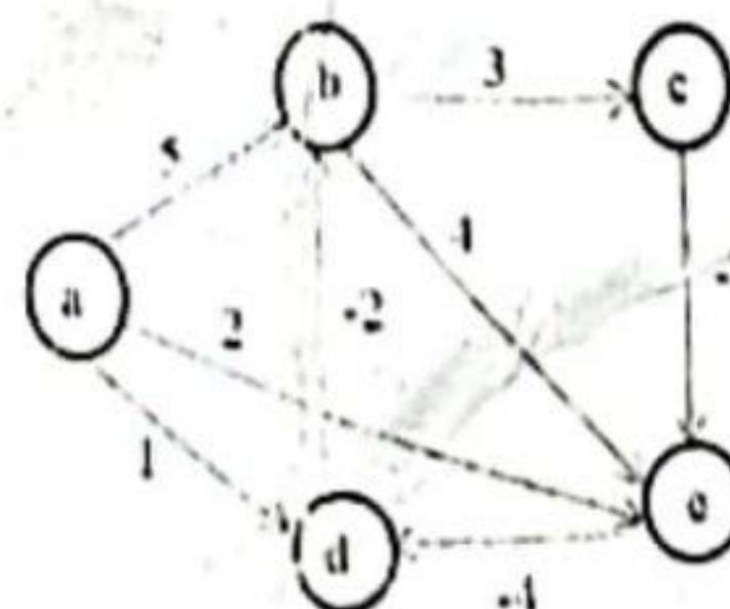
5. a) Considering the following undirected weighted graph, find the shortest path between the node 0 and 5. Clearly show the interim steps of your process. 08



b) Design the algorithm/pseudo code of the above response 04

6. a) Discuss the pros and cons of 'Floyd Warshall algorithm' over the other shortest path finding algorithms. Explain a case where 'Floyd Warshall algorithm' will fail to find solution. 3+3

b) Identify the shortest path between a and e using Floyd Warshall algorithm. Clearly demonstrates the steps 06



(6) (a) Define Hermitian and Skew Hermitian matrices. Show that every square matrix may be written as the sum of a Hermitian and a skew-Hermitian matrices. 06

(b) Show that the transpose of the product of two matrices is the product in reverse order of their transposes i.e., $(AB)' = B'A'$. 03

(c) Show that the both matrices $\begin{pmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 4 & 3 & 3 \\ -1 & 0 & -1 \\ -4 & -4 & -3 \end{pmatrix}$ are involutory. 03

(7) (a) Define upper triangular and lower triangular matrices, symmetric and skew-symmetric matrices and orthogonal matrix. 05

(b) Find the inverse of the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{bmatrix}$

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ B_{21} & B_{22} & B_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

Also find the symmetric and skew symmetric parts of the above matrix A . 07

(8) (a) Define minor of a matrix. Find the rank of the matrix using minor process

$$A = \begin{bmatrix} 0 & 6 & 6 & 1 \\ -8 & 7 & 2 & 3 \\ -2 & 3 & 0 & 1 \\ -3 & 2 & 1 & 1 \end{bmatrix}$$

(b) Find the rank of the matrix $A = \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ by reducing it to canonical form. 07

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\begin{aligned} a &= -2 \\ b &= 1 \\ c &= -3 \end{aligned}$$